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Technical and economic feasibility of a 50 MW grid-connected solar PV at UENR Nsoatre Campus

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ABSTRACT

The purpose of this study is to investigate the technical and economic feasibility of a 50 MW grid-tied solar photovoltaic plant at UENR Nsoatre Campus. The suitability of the site for PV plant development is initially examined with site assessment criteria. PVsyst software is then used to model, simulate and estimate the performance of three PV technology plants. Economic analysis is conducted on the three PV systems with RETScreen software. NASA SSE solar irradiation data set from NASA website were used for the assessments. Site assessments results showed that mono-crystalline silicon, poly-crystalline silicon, and thin film (CdTe) systems would occupy 9%, 10%, and 13% respectively of the 2000 acres acquired for the campus. The performance analysis showed that monocrystalline silicon, poly-crystalline silicon and thin film (CdTe) systems would export 67315 MW h, 67506 MW h, and 68991 MW h respectively to the grid annually while meeting more than 48% of campus electricity needs. The costs of the energy produced by the systems are 12.4 cents/kWh, 12.3 cents/kWh and 10.9 cents/kWh for mono-crystalline, polycrystalline and thin film systems respectively. The determined costs of energy for the systems are lower than the set feed-in tariff of 14 cents/kWh. NPV was positive for all the systems while; benefits to cost ratios were greater than 1.0 with reasonable payback periods of 7.2 years for crystalline silicon technology systems and 6.4 for thin film technology system.

1. Introduction

The power sector in Ghana, like many others in sub-Saharan Africa, is facing a number of challenges. The perennial low water levels of Ghana's hydroelectric dams occur due to variability in rainfall patterns and are among the main challenges of the sector. In 2016, a 33% drop in total generation output of the installed hydro dams in Ghana was due to low water levels (Energy Commission, 2017). These hydro dams account for more than 41% of the total installed electricity generation capacity in the country (Kumi, 2017). The demand for electricity has been steadily increasing, at an estimated 12% per annum due to expanding economy, rapid urbanization, industrialization and rural electrification (Gyamfi et al., 2017). These challenges have resulted in power supply deficits, which have accounted for past perennial power outages. It is estimated that Ghana loses between 2% and 6% of its annual gross domestic product (GDP) due to inadequate and unreliable power supply (Eshun and Amoako-Tuffour, 2016).

The government's efforts to address the power supply deficit have been towards capacity additions, which are mainly based on fossil fuels and demand high capital investments. Karpowership and the Electricity Company of Ghana (ECG) signed a power purchase agreement in June 2014. The purpose of the agreement was to help address a power supply challenge that has resulted in loading scheduling. The AMERI Power Purchasing Agreement was also signed in 2016 to provide 300 MW of emergency power to manage the debilitating energy crisis of that year (Gyamfi et al., 2017). Power plants based on fossil fuels now represent more than 57% of the total electricity generation capacity (Kumi, 2017). The use of these power plants to address the electricity supply deficit has also been restrained by fuel supply constraints. This has adversely affected the power generation of the installed thermal power plants (Sakah et al., 2017).

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Nomenclature

CdTe	Cadmium Telluride
CI (G) S	Copper Indium Gallium Selenide
ETU	Electricity Transmission Unit
FiT	Feed-in Tariff
KWh/m2/day	Kilowatt per square meter per day
Lc	Normalized Array Losses
LCOE	Levelized Cost of Electricity
Lcr	Array loss/Incident Energy Ratio
Ls	Normalized System Losses
Lsr	System Loss/Incident Energy Ratio
MWh/yearm2	Megawatt hour per year per square meter
NITS	National Interconnected Transmission System
OPEX	Operational Expenditure
PR	Performance Ratio
RETs	Renewable Energy Technologies
Ya	Normalized Array Production
Y_f	Normalized System Production
Y_r	Reference Incident Energy in Coil Plane

On the other hand, there are plentiful renewable energy resources in Ghana, such as biomass, solar, and wind energy resources, that can be utilized for the socio-economic development of the country (IEA, 2015). With the support of targeted policies, the use of these renewable energy resources can promote energy access, particularly in rural areas. The conversion of these resources into electricity can also improve energy security and improve socio-economic development (International Renewable Energy Agency, 2015). Moreover, renewable energy deployment can not only help reduce the government's dependence on fossil fuel power plants but also help mitigate climate change (Gielen et al., 2019).

Globally, solar photovoltaic has become an attractive alternative to governments for power infrastructure expansion due to the increasing technological advancements and decreasing costs of photovoltaic (PV) systems (Duan et al., 2013). In Ghana, however, it has been identified that a key barrier to the development and deployment of renewable energy resources is the low level of research on local renewable resources (Ben Hagan, 2015).

Due to the variable output profiles of renewable technologies, it is important to properly assess the specific local conditions to ensure the reliability of such energy supplies. Furthermore, the investors' willingness to commit capital to renewable energy projects are majorly driven by the risk return profiles of such investments. In this context, the primary objective of this study is to investigate the technical and economic viability of implementing a 50-MW grid-connected PV system at the Nsoatre Campus of the University of Energy and Natural Resources (UENR).

The present study aspires to contribute to the much needed research in the renewable energy sector in Ghana and help facilitate the development and deployment of renewable energy in the country.

2. Materials and methods

The methodology adopted to perform the technical and economic feasibility study of the 50-MW solar PV plant is a three-phase approach, as illustrated in Fig. 1.

Firstly, the pre-feasibility phase begins with a brief description of the project site and characterization of the new campus' electricity requirements. The pre-feasibility, which is part of the technical assessments, involved the assessment of local solar resources.

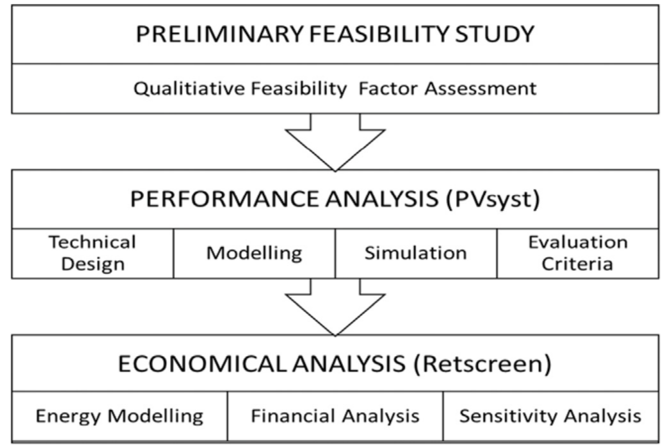

Fig. 1. Methodology flow chart.

The feasibility factors reviewed at this stage include assessment of transmission and distribution infrastructure, access infrastructure, and land availability. Site visits, literature surveys, and the PVsyst software were used as tools to model and estimate the energy yields of the available solar resource.

The second stage was the performance assessment. This stage involved technology specific information, including types of technologies, costs, efficiencies, and land requirements. Solar radiation data were combined with technology specific information for the energy model. The PVsyst software was used to develop an energy model for the assessment.

After the system was designed and the performance was analysed based on of the various PV system options, the economic analysis of each system was conducted. The RETScreen software was used to model the financial aspects of the various PV systems. The results include details on capital and operating cost requirements of each PV system.

2.1. PV system simulation tools

RETScreen and PVsyst were chosen for the simulation and analysis due to their combined comparative advantage. While RETScreen offers superior tools for cost, financial risk, and emission analyses, PVsyst provides tools to assess technical details of grid-connected PV systems. The combined strengths of these software provide the required details to assess the project viability factors considered in this study. RETScreen has sufficient and validated results in addition to a wide range of integrated data regarding power plants and meteorology, which assure the accuracy and quality of results (Kolhe et al., 2015). Moreover, PVsyst is widely used due to its flexibility and accuracy in modelling large scale PV plants (International Renewable Energy Agency, 2015). PVsyst also provides tools to perform a wide range of calculations including detailed losses (ohmic losses, soiling loss, shading, etc.) which are not available in RETScreen (Calvert and Simandan, 2010).

2.2. Grid-connected PV system performance indicators

Performance indicators of grid-connected PV plants are used in the assessment of PV projects and are important to assess the viability of solar projects. These indicators provide the basis to calculate the project revenue. They are also used to predict the average annual energy output for the lifetime of the proposed power plant, which is typically 25–30 years. Indicators are used to evaluate the overall performance of grid-tied PV systems and can be used to compare the performance of different PV configurations or systems (Schweiger et al., 2017).

The most common performance indicators that define the overall system performance with respect to energy production, solar resource, and overall effect of system losses are final PV system yield, reference yield, performance ratio, capacitor factor, and specific yield (Miller and Lumby, 2012).

2.2.1. Final PV system yield (Yf)

The final PV system yield, Y_f, is defined as the net energy output E divided by the nameplate D.C. power, P_O, of the installed PV array. Y_f represents the number of hours that the PV array would need to operate at its rated power to provide the same energy. The units are in hours or kWh/kW. Y_f normalizes the energy produced with respect to the system size. Consequently, it is a convenient way to compare the energy produced by PV systems of different sizes..

$$Y_f = E/_{P_o} \quad \left(\mathrm{kWh/kW}\right) \tag{1}$$

2.2.2. Reference yield (Y_r)

The reference yield, Yr, is the total in-plane irradiance, H, divided by the PV's reference irradiance, G. It represents an equivalent number of hours for the reference irradiance. If G equals 1 kW/m2, then H is the number of peak sun hours or the solar radiation in units of kWh/m2. Yr defines the solar radiation resource for the PV system. It is a function of location, orientation of PV array, and monthly/yearly weather variability.

$$\boldsymbol{Y_r} = {}^{\boldsymbol{H}}\!/\!_{\boldsymbol{G}} \tag{2}$$

2.2.3. Performance ratio (PR)

The performance ratio, PR, is the Y_f divided by the Y_r. By normalizing with respect to irradiance, PR quantifies the overall effect of losses on the rated output due to: inverter inefficiency, wiring, mismatch, and other losses when converting DC to AC; PV module temperature; incomplete use of irradiance due to reflection in the module front surface; soiling or snow; system down-time; and component failures.

$$PR = Y_f \Big/ Y_r \tag{3}$$

2.2.4. Specific yield

The specific yield (kWh/kWp) is the ratio of total annual energy generated (kWh) to the installed capacity (kWp). It determines the financial value of PV plants and can be used to compare the results from different PV technologies and systems. The specific yield of a PV system is given by Eq. (4)

$$S_y = \frac{Energy\ generated\ per\ anum\ (kWh)}{Installed\ capacity\ (kWp)} \tag{4}$$

2.2.5. Capacitor factor (CF)

The capacity factor (CF) of a PV power plant is defined as the ratio between the output over a year and the output if it had operated at nominal power that entire year. CF is given in Eq. (5).

$$CF = \frac{Energy\ generated\ per\ anum\ (kWh)}{8760\left(\frac{hours}{anum}\right) xInstalled\ capacity\ (kWp)} \tag{5}$$

$$CF = S_y/8760 \tag{6}$$

The specific yield of a PV system is related to CF by the expression in Eq. (6). The CF of a fixed tilt PV plant can vary from 12 to 24%, depending on the solar resource and the performance ratio of the plant (Miller and Lumby, 2012).

2.3. Economic evaluation criteria of solar PV systems

Solar PV systems are generally characterized by high fixed cost and low operation cost. Below is a discussion of key economic criteria for solar PV investments evaluation.

2.3.1. Levelized cost of electricity (LCOE)

The levelized cost of electricity (LCOE) gives a convenient summary measure of the overall competitiveness of different electricity generating technologies. It presents the total cost of installing and operating a project expressed in dollars per kilowatt-hour of electricity produced by the system over its lifetime (AlAjmi et al., 2016). It takes into account installation costs, financing costs, taxes, operation and maintenance costs and the quantity of electricity the system generates over its life. The following expression is used to calculate LCOE (Allouhi et al., 2016)

$$LCOE = \frac{\sum_{t=0}^{LP} C_t \Big/ (1+r)^t}{\sum_{t=0}^{T} E_t \Big/ (1+r)^t} \tag{7}$$

Where;

LP is the life of the project [years], t is the year, and r is the discount rate for t (%). Ct and Et are the net cost of project for t [$] and energy produced for t [kWh]. Ct and Et can be expressed by Eq. (8).

$$C_t = I_t + O_t + M_t : \tag{8}$$

$$E_t = S_t(1-d)^t \tag{9}$$

In the two previous equations, I_t is the initial investment of the system, including construction, installation and other expenditures. O_t and M_t are the operation and maintenance costs for t, respectively. S_t expresses the yearly rated energy output [kWh] and d is the degradation rate of the PV panels [%].

2.3.2. Payback period (PBP)

Payback period is defined as the length of time for an investment to recover its costs. PBP is the time required for the accrued revenue to be equal to the total initial investments. Though PBP typically ignores the discounted revenue earned, it is an important determinant of how long a project may take to recover.

2.3.3. Net present value (NPV)

NPV is the discounted sum of the revenue from selling the generated energy net of all costs associated with the energy delivery system. This criterion takes into account the time value of money and the useful life of the project. All anticipated costs are discounted to the present time, and are termed as the present worth of the cost. The NPV is the sum of all the present-worths, where present worth Ŋ′ of $ at k years in the future, for a market discount rate ɑ, can be calculated as: Eq (10)

$$\mathfrak{y}' = \frac{p}{(1+\mathfrak{a})^k} \tag{10}$$

Apart from the market discount rate, the recurring future cash flows might be assumed to inflate (or deflate) at a fixed percentage g. Hence, the worth of \$P at the end of the *k*th year would be greater than \$P, and equal to $\$P(1+g)^{k-1}$. Furthermore, the present worth Ω'' after considering the inflation rate g of \$P at the end of year k can be given as

$$\Omega'' = \frac{p(1+g)^{k-1}}{(1+a)^k} \tag{11}$$

Consequently, the sum of the present-worths of all the anticipated future savings would yield the NPV. NPV is the most appropriate economic criteria for solar PV investment analysis as it incorporates the entire life-cycle of the projects and hence the time value of money (Kornelakis and Koutroulis, 2009). Conventionally, NPV is calculated for all the proposed projects, and the project with the highest NPV is selected.

2.3.4. Return on investment (ROI)

Return on investment (ROI) is defined as the ratio of net profit to cost of investments expressed as percentage. ROI is a measure of the gain or loss accrued on an investment based on the amount of money invested. It is a decision tool for comparing the profitability or the efficiency of different investments

$$ROI = \frac{NET\ PROFIT}{Cost\ of\ Investments}\ x100 \tag{12}$$

2.4. Description of the proposed UENR Nsoatre Campus

The proposed site is located in the Sunyani West District of the Brong Ahafo Region. The coordinates for the location are 7°24′38″ N and 2°28′58.9″ W. Its northern part lies along the Trans West African Highway (N6). The Tain Block II and Asukesi forest reserves are located in the northern and southern parts, respectively, as shown in Fig. 2. In the south eastern side of the site, lies in the Nsoatre Township. Currently, as a virgin site with agricultural land use, the site is 18.7 km away from Sunyani and 15 0.1 km from Berekum.

2.5. Electricity consumption data of UENR

Three years data on the energy consumption of the University campus was gathered from the University's accounts office and further processed for use in the project. Based on the obtained data, 660,200 kWh, 694,373 kWh, 732,293 kWh of electricity were consumed in 2015, 2016, and 2017, respectively. The average monthly consumption for the period ranged between 39,948 kWh in August and 70,773 kWh in October. There are two peak consumptions periods in the year. The first occurs between March and May, and the second in October, as shown in Fig. 3. These peak periods represent the two semesters of the year. The first is usually between January and May, and the second between September and December. Between these two periods, there are semester breaks, where the power consumption on campus is reduced.

2.6. Site assessment criteria and methodology

The purpose of this stage is to identify basic project implementation barriers. Site assessment criteria were employed to identify barriers for the viability of the project. Visits and literature surveys were employed to assess the site. The viability factors considered in this study include solar resource, land availability and topography, infrastructure, local climate, and financial incentives.

2.6.1. Solar resource assessment

PV systems outputs are directly linked to the amount of solar irradiation at the selected location. A number of factors influence the availability of solar resource at a particular location. This includes; latitude of the location, season and time of the day, orientation, and surface inclination. As noted by Palmer et al. (2017), solar resource is strongly affected by long-term weather patterns (cloud cover), atmospheric conditions (such as dust and pollution levels) as well as the physical characteristics and topology of the Earth's surface.

2.6.2. Land availability and topography

Slope and soil types require proper assessment since levelling and terracing works can be costly (Miller and Lumby, 2012). In addition, sufficient area must be available for the required capacity to avoid pitch reductions that may result in yield losses (Miller and Lumby, 2012). Ideal sites for PV projects are levelled grounds with south-facing gentle slope north of the equator, or north-facing gentle slope south of the equator. These conditions lead to simpler installations and reduced costs (Arán Carrión et al., 2008).

2.6.3. Local climate

Climate conditions affect the performance of PV plants. Extreme weather conditions can increase the risks of equipment damage. Therefore, it is essential to investigate weather conditions such as temperature, rainfall, and high wind speeds (Hasapis et al., 2017; Miller and Lumby, 2012; Quansah et al., 2017).

2.6.4. Infrastructure

Regarding accessibility, the selected area should have access roads for the delivery of PV systems and other construction materials. The costs to construct new roads or improve existing ones are considerably reduced if they are close to main access road (<1 km or 1–2 km) and may lead to lower costs in project investments (Fernandez-Jimenez et al., 2015).

Factors that affect grid connection include grid capacity, proximity, right-of-way, and grid stability (Miller and Lumby, 2012). It is desirable that the PV plant is somewhat close to the existing grid infrastructure to reduce grid connection costs. Additionally, the existing grid infrastructure should have sufficient capacity to accommodate the power generated by the PV plant.

2.6.5. Financial incentives

Public incentives are one of the major drivers of the global growth of PV energy (Jaeger-Waldau, 2017). The existence of financial incentives such as feed-in tariffs (FiT) and tax breaks can greatly influence the viability of PV projects in developing countries due to the required high capital investments. Though financial incentives differ from country to country, the economic benefits of such incentives could exceed the constraints of one or two site selection viability factors (Miller and Lumby, 2012). The political and regulatory risks should be thoroughly assessed to ensure that the financial incentive will remain in force in countries where they exist.

3. Plant design criteria

To achieve optimum balance between system performance and costs, the proposed plant requires careful selection, sizing, and matching of the system components. Therefore, a thorough

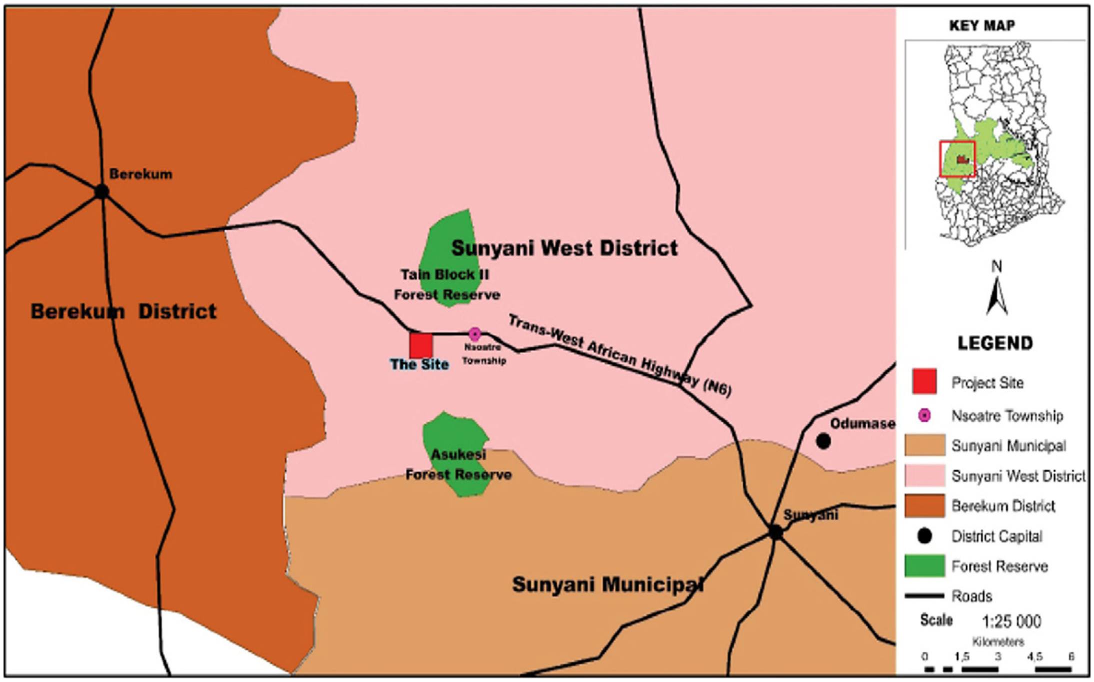

Fig. 2. Location of the Project site.

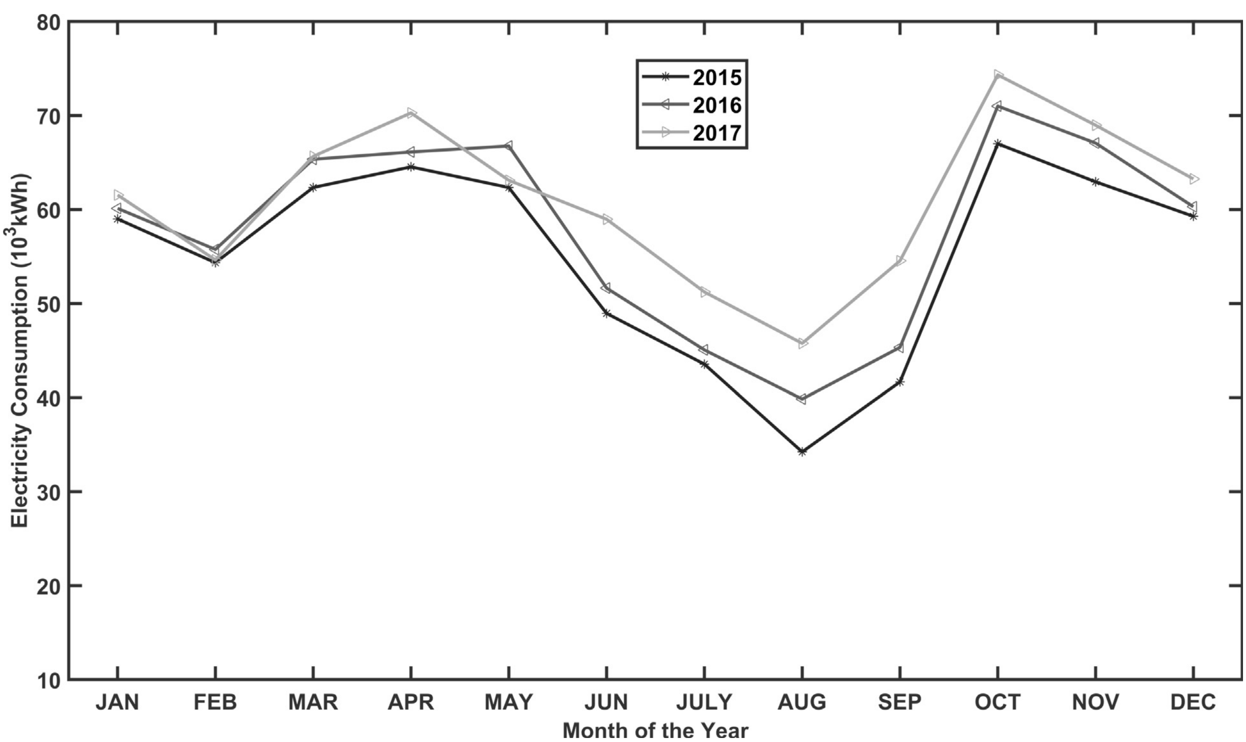

Fig. 3. Monthly electricity consumption in the UENR Sunyani campus.

investigation of the main design aspects to achieve the desired design objectives should be performed (Hasapis et al., 2017; Miller and Lumby, 2012). The design aspects include the selection of the key technologies, layout, and characteristics of the electrical system.

3.1. Technology selection

The selection of PV technologies was based on the commercially available PV products, including first and second generation technologies. These commercially available technologies are evaluated in subsections 3.1.1-3.1.2. Additionally, the balance of system components, including inverter and mounting structures, was also evaluated for the design.

3.1.1. Solar photovoltaic modules

The cost-effectiveness of a PV system depends on the cost of the selected modules. It is important to examine the long term cost implications of the selected modules with respect to their performance. For instance, crystalline cell technology is the most efficient technology available today. However, less efficient technologies such as CdTe offer better cost performance index (Horowitz et al., 2017). The selection of PV modules was based on the cost, electrical characteristics, spectral response, degradation, warranty, and lifetime analyses.

3.1.2. Inverters

Inverters are key components of grid-connected PV systems. They convert the DC power from the PV arrays to AC power that matches the grid power in terms of phase and voltage. The inverter for a grid-connected PV system should be meticulously selected to maximize reliability and quality throughout the operational life cycle. Incorrect sizing of inverters can result in load mismatches that reduce the performance of the PV system.

3.1.3. Mounting structures

It is important that PV modules are mounted on stable structures so that they remain correctly oriented and provide structural support and protection for the modules. Fixed tilt mounting systems are simpler, cheaper and require lower maintenance compared to tracking systems.

3.2. Layout

PV plant layout designs depend on the site conditions. The accessible space and geographical topology of the site can be a constraint to the desired layout plan. Nevertheless, the objective of a good layout design is to realize maximum energy output from the PV plant while minimizing costs.

3.2.1. Tilt angle

An optimal tilt angle is required for maximum yield in a PV system of a given location. The optimal tilt angle can be determined with a simulation software or through calculations using the location's latitude (Miller and Lumby, 2012). An optimally determined tilt angle must be reasonably adjusted considering soiling and shading for optimal performance.

3.2.2. Orientation

The ideal optimal azimuth of a particular PV project can be determined using the PVsyst software. The azimuth of a PV system directly impacts the output yield of the system. Therefore, the azimuth should be optimally determined to avoid possible losses.

3.2.3. Inter-row spacing

The spacing between rows of arrays is limited by the land size and inter-row shading. While it is important to allow sufficient spacing between rows of arrays to avoid inter-row shading, long cable runs should be avoided. These measures can reduce possible ohmic losses and further ensure the judicious use of available land.

3.3. Electrical design

The electrical design of the systems comprises the design of direct current (DC) and alternating current (AC) systems of the plant. Design details of the PV array and inverter are discussed in subsections 3.3.1 and 3.3.2.

3.3.1. PV array sizing

The sizing of the PV array is based on the inverter specifications and the system architecture. In most cases, a sizing ratio between 1 and 2 maximizes the energy produced per system. Other considerations include optimization of the system voltage to achieve the maximum efficiency of the inverter. For that, the voltage of the array close to the inverter optimum voltage should be fixed based on inverter voltage dependency efficiency graphs (Hasapis et al., 2017). Safety requirements with respect to inverter voltage limits should be consistent with national regulations.

3.3.2. Inverter sizing

Optimal PV/inverter sizing depends on factors such as local climate, PV surface orientation and inclination, inverter performance, and PV/inverter cost. When the PV array generates power that is only a fraction of its rated capacity under low insolation conditions, the inverter operates under partial load with lower system efficiency. The efficiency of the PV systems drops when the inverter's rated capacity is significantly lower than the rated capacity of the PV system and, consequently, the inverter becomes overloaded. Under overloading conditions, the excess PV output, which is greater than the inverter rated capacity, is lost. Therefore, over or under sizing increases the costs as noted by Mondol et al. (2006). The optimum sizing or power ratio for a high efficiency inverter system at high or low insolation locations should be within 1.1–1.2 and 1.3–1.4, respectively. For a low efficiency inverter system, the optimum sizing should be within 1.2–1.3 and 1.4–1.5 (Mondol et al., 2006).

3.4. Proposed PV systems specifications

This study considered three different PV technologies for the design of the proposed 50-MW PV system: mono-crystalline silicon (mono-Si), poly-crystalline silicon (poly-Si), and cadmium telluride (CdTe) from thin film technology. The specifications of the selected PV modules at standard test conditions are listed in Table 1.

An SMA Sunny Central 2500-EV inverter was selected for the proposed systems. This inverter produces 2500 kVA from 1500 V DC and allows for an efficient system design as it works with a broad range of module types. It has an integrated transformer, and it is optimized for outdoor installation. Table 2 shows a summary of the design specifications of the three PV systems analysed.

3.5. Simulation and evaluation criteria

The inputs to the modelled designed systems consisted of information on solar resource and temperature conditions of the site, in addition to the layout and technical specifications of the plant components. To determine the specific performance indices of the analysed PV systems, one year hourly simulations were performed using the PVsyst software version 6.66.

3.6. Economic evaluation

The relative cost effectiveness of the various systems were estimated and compared using key economic evaluation criteria. The main issues affecting the economic evaluation of the PV systems included capital expenditure (CAPEX), operational expenditure (OPEX), and cost of produced energy. Additionally, the timing of expenditure and revenue and the financing costs were also evaluated.

3.6.1. Methodology

The first step in this phase is the determination of the project costs. Project capital costs, operational costs, and financing costs

Table 1
Technical Specifications of the PV modules.

PV Module Technology	Mono-si	Poly-si	Thin Film (CdTe)
Manufacturer	Jinkosolar	Jinkosolar	First solar
Model	JKMS360	JKM 350P	FS-4120-3A
Nominal power (W_p)	360	350	120
Open-circuit voltage - V_{OC} (V)	48	48	88.7
Short-circuit current - I_{SC} (A)	9.51	9.36	1.84
Voltage at point of maximum Power-V_{MPP} (V)	39.5	38.6	70.7
Current at Point of Maximum Power- I_{MPP} (A)	9.12	9.07	1.695
Module efficiency - η_m (%)	18.6	18.07	16.66
NOCT (^oC)	25	25	25
Temperature coefficient (mA/^oC)	4.7	5.6	0.7
Module Area (m^2)	1.94	1.952	0.72

Table 2
Summary of design specification of the PV systems.

PV System	Mono-Si	Poly-Si	Thin Film (CdTe)
Module Type	JKMS360M	JKM 350 PP-72-DV	FS-4120-3
Total No. of Modules	140832	142857	41668
Modules in Series	25	27	14
Modules in Parallel	5556	5291	29762
Inverter Capacity (kW)	42500	42500	42500
System Capacity (kW)	50004	50000	5000

were estimated using data from the literature and industry. Subsequently, the project revenues inflows were estimated. RETScreen was used to model and analyse three technologies (monocrystalline, polycrystalline, and thin film PV systems) to assess their relative cost effectiveness.

3.6.2. Project costs components

There are wide variations in the costs for different market segments and countries. The cost of a PV system depends on the system size and other factors, such as geographic location, mounting structure, and type of PV module (Feldman et al., 2012). The size of a project has significant effects on the cost of investments due to economies of scale; the larger the project, the lower the investment costs. The cost stream constitutes three components: CAPEX, which comprises investments covering all costs associated with the designed PV systems; OPEX, which is associated with the systems; and the financing costs.

3.6.3. Capital expenditure

CAPEX includes the cost of purchasing all components, the installation and civil work costs, feasibility study cost, and engineering and development costs. The initial investment represents the price of photovoltaic panels, mounting structure, inverters, cabling, disconnects, and metering devices. Conservative estimates of \$0.71/Wdc, \$0.68/Wdc, and \$0.58/Wdc were calculated for mono-Si, poly-Si, and thin film (CdTe), respectively. The rest of the project's CAPEX including BOS were based on a study by the International Renewable Energy Agency (IRENA) (Renewable Energy AgencyI, 2016).

3.6.4. Operation expenditure

The main maintenance cost is the yearly cost of cleaning the solar arrays and other maintenance tasks such as replacing cabling or performance check-ups. The cost is determined by the price and quantity of man-hours attributed to the systems under consideration. The operating and maintenance costs were obtained by the sum of the costs of replacement of malfunctioning panels, replacement of inverters, and maintenance of the system. The OPEX estimates were based on studies by IRENA (Renewable Energy AgencyI, 2016).

3.6.5. Financing costs

It was assumed that a percentage of the total project cost would be financed by loans. Therefore, the cost of financing was added to the overall project costs. The financing costs comprise annual payments of interest at a rate of 5% for 10 years.

3.6.6. Revenues

The revenue stream is mainly the revenue from the sale of generated electricity to the utility company. The rate at which electricity is sold is based on the prevailing feed-in-tariff (FiT) of 14 cents/kwh with ten years guarantee. For the purposes of this analysis, the project's lifetime is 25-years.

4. Results and discussion

4.1. Technical analysis

The site assessment is discussed based on a thorough analysis of the simulation results of the designed photovoltaic systems.

4.2. Summary of qualitative factor assessments

4.2.1. Solar resource assessment

The data obtained from the Surface Meteorology and Solar Energy programme (NASA-SSE) (Fig. 4) indicates that the entire site

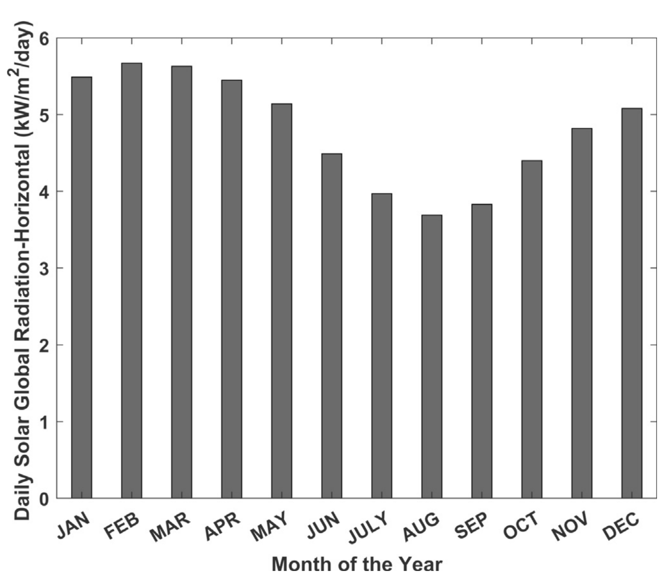

Fig. 4. Daily solar radiation at the UENR Nsoatre campus.

for the campus experiences an average of 1752 kWh/(m^2·Yr) of solar irradiation throughout the year. The monthly average irradiation on a horizontal surface ranges from 3.97 to 5.67 kWh/(m^2·day), with an average of 4.81 kWh/(m^2·day). This result is approximately 70% higher than the monthly average irradiation for Germany, the world's leader in grid-connected PV with over 40 GW of installed capacity (Ritter et al., 2016).

4.2.2. Accessibility

The Trans West African Highway (N6) offers an excellent road network that connects the proposed site to the Sunyani and Berekum Municipal assemblies. The construction of primary and secondary roads from the Trans West African Highway (N6) to the campus as proposed in the master plan will offer access to the project site. Road access will not be a constraint to the development of the project and may not affect project investments cost.

4.2.3. Utility grid connection proximity and capacity requirements

The electricity generated from the plant can be fed into the grid through the 161 kV transmission line from Sunyani to Berekum. The transmission line is located less than 200 m from the project location.

4.2.4. Land topography and acreage requirements

Fig. 5 shows a topographic map of the campus with increasing elevation from north to south. The upper area of the southern part of the campus land is mostly constituted of scattered chamber systems that are almost buildings. However, it will require adequate levelling to avoid possible shadings and to help achieve the optimum tilt angle for the solar panels.

The results show that land requirements vary depending on the type of PV technology. To calculate the required area, solar panels were assumed to be placed further apart to minimize losses due to shading, thus maximizing the generation potential of the systems. To achieve this objective, calculations were based on the results of a study by Ong et al. (2013). Monocrystalline, polycrystalline, and thin film would occupy 9%, 10%, and 13% of the 2000-acres campus.

4.3. PV system performance optimization results

The performance indices discussed in this study are the results of the modelling and simulations performed using the PVsyst 6.66.

4.3.1. The final yield and net energy output

Fig. 6 shows the plot of the average final yield (Y_f) generated by the PV systems in year one. The mono-Si system recorded the lowest final yield of 2.66 h in August, followed by the poly-Si system with 2.67 h, and CdTe with 3.02 h. CdTe recorded the maximum final yield of 4.81 h in January, followed by poly-Si with 4.69 h, and mono-Si with 4.68 h. These recorded values compare favourably with a study on an existing grid-connected system at the Kwame Nkrumah University of Science and Technology (KNUST) in Kumasi (Quansah et al., 2017), where the obtained final yields were 2.89 h, 3.1 h, and 3.08 h for mono-Si, poly-Si, and amorphous-Si, respectively. However, the obtained final yields were much lower than the values of similar studies performed by Adaramola (2014), who obtained 4.58 h for a poly-Si system, and Allouhi et al. (2016), who obtained 4.85 h and 4.98 h for mono-Si and poly-Si systems, respectively.

Fig. 7 shows the total annual energy generated by the PV systems in the first year. The total annual energy delivered to the grid in year one varied between 67,315 MWh for mono-Si, 67,506 MWh for poly-Si, and 68,991 MWh for thin film (CdTe) technology.

The highest final yield (7464 MWh) was recorded in January, and the lowest (4212 MWh) in August. The crystalline silicon technology delivered noticeably lower average monthly energy to the grid compared to the thin film (CdTe) technology. However, poly-Si presented slightly higher values than mono-Si technology (Fig. 7).

4.3.2. Performance ratio (PR)

The results showed that CdTe is the most efficient system and converted more than 77% of the solar resource into useful energy. CdTe was followed by the poly-Si system with a 75.7% PR, which is slightly above the 75.5% PR of the mono-Si system. The obtained PR of the mono-Si system is 10% higher compared to the 67.9% obtained by Quansah et al. (2017) on an existing grid-tied system.

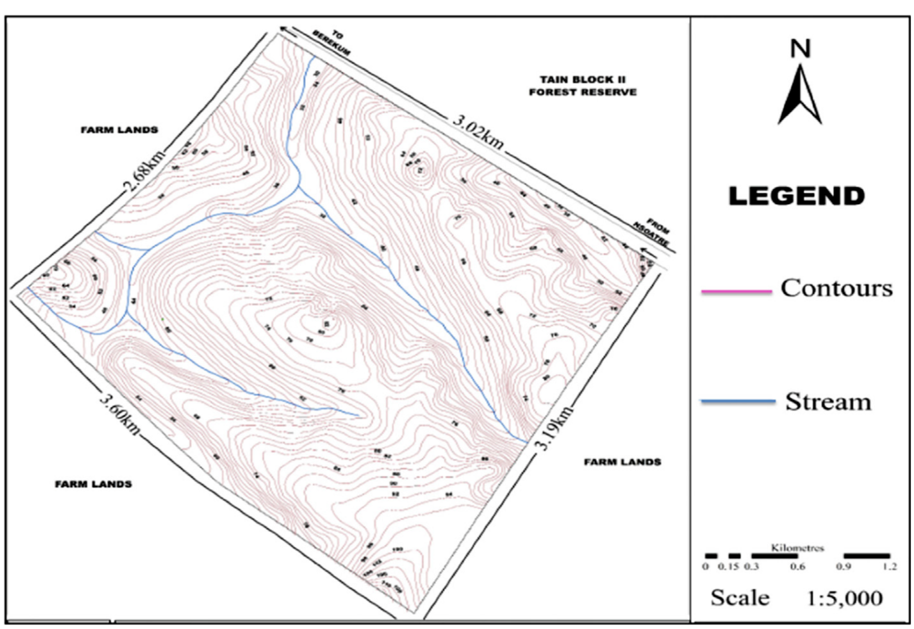

Fig. 5. Topographic map of site.

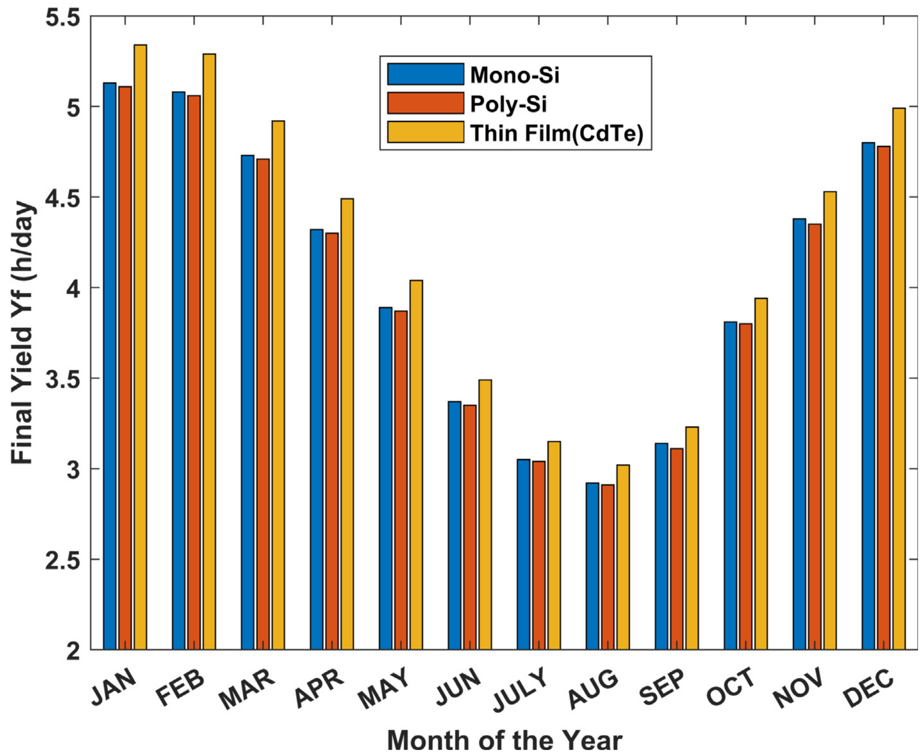

Fig. 6. Comparison of Monthly Final Yield Yf of the systems.

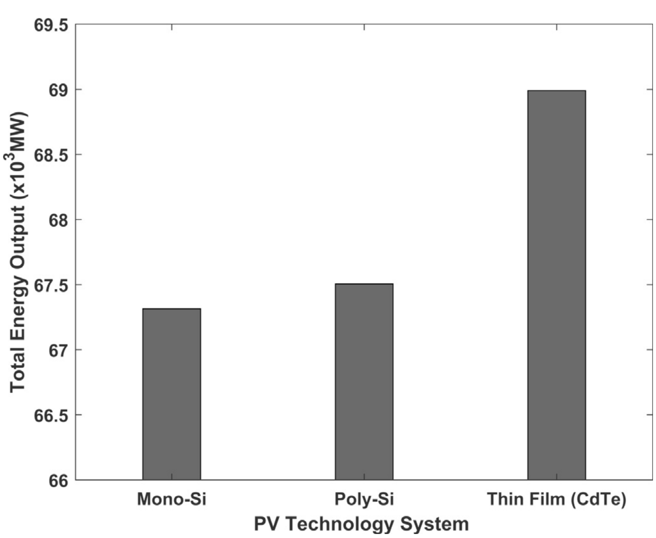

Fig. 7. Actual annual energy generated by the systems in year one.

However, the PR of poly-Si system is 6% lower than the PR of an existing grid-tied system analysed by Quansah et al. (2017). The differences in the PR values of simulated systems and the actual grid-tied system can be attributed to unaccounted losses due to inverter inefficiencies, wiring, mismatches, soiling, system downtime, and component failures. Fig. 8 shows the average PR trend in the first year.

4.3.3. Total energy losses

It can be observed from Fig. 9 that crystalline technology systems recorded the highest losses compared to the thin film (CdTe) technology. These losses constitute 15%, 13%, and 9% of the final energy delivered to the grid by poly-crystalline, mono-crystalline, and thin film technologies, respectively. It is observed from Fig. 9 that crystalline silicon technology systems recorded the highest losses compared to the Thin Film technology.

4.3.4. The annual capacity factor

The CF calculated for the simulated systems are 15.37%, 15.41%, and 15.75% for mono-Si, poly-Si, and thin film (CdTe) technologies, respectively. These values are comparable to actual grid-tied PV installations, as shown in Table 3.

4.4. Economic analysis

4.4.1. Results and discussions

The proposed systems can provide, on average, 48% of the total monthly electricity demand of the University Campus. This value is rather low because the electricity consumption data were daily averaged instead of hourly. A study by Kawamoto et al. (2000) observed that 65% of office equipment are turned off at night, while the remaining 35% are active. This suggests that the greater portion of electricity consumed on the university campus is consumed in the day. Fig. 10 shows the total electricity demand on campus, the power supplied by the systems, and the projected supply. According to the analysis, approximately $305,234 could be saved annually if all energy demand was supplied by PV systems. It can be deduced from this graph that there are several opportunities to optimize the power supply in the proposed campus. For instance, high electricity consuming appliances may be operated in the day to make use of available power from the PV plant.

The initial costs of the mono-Si, poly-Si, and CdTe PV systems were estimated at $69.5 million, $68.1 million, and $65.5 million, respectively. The higher cost of the mono-Si system costs is primarily due to the high cost of monocrystalline PV modules. Table 4 shows the summary of the economic indicators of the systems. The

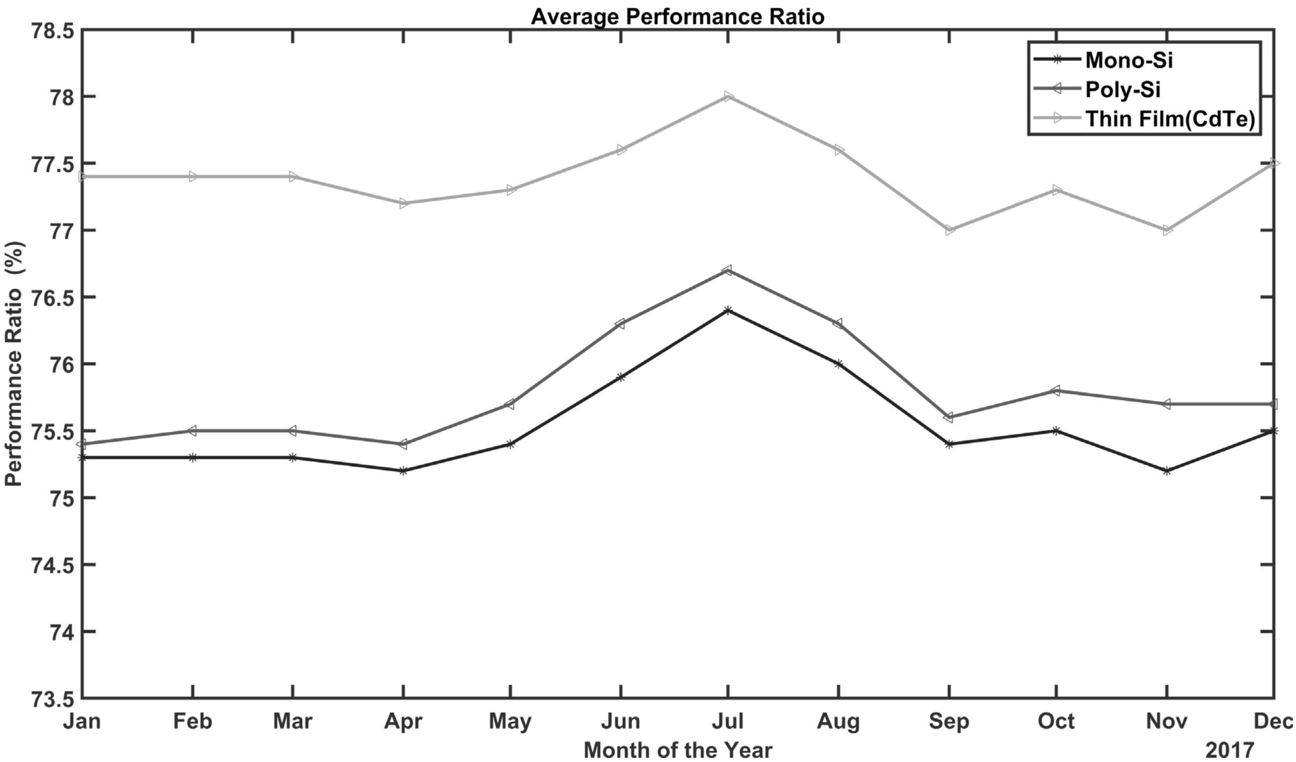

Fig. 8. Average performance ratio trend.

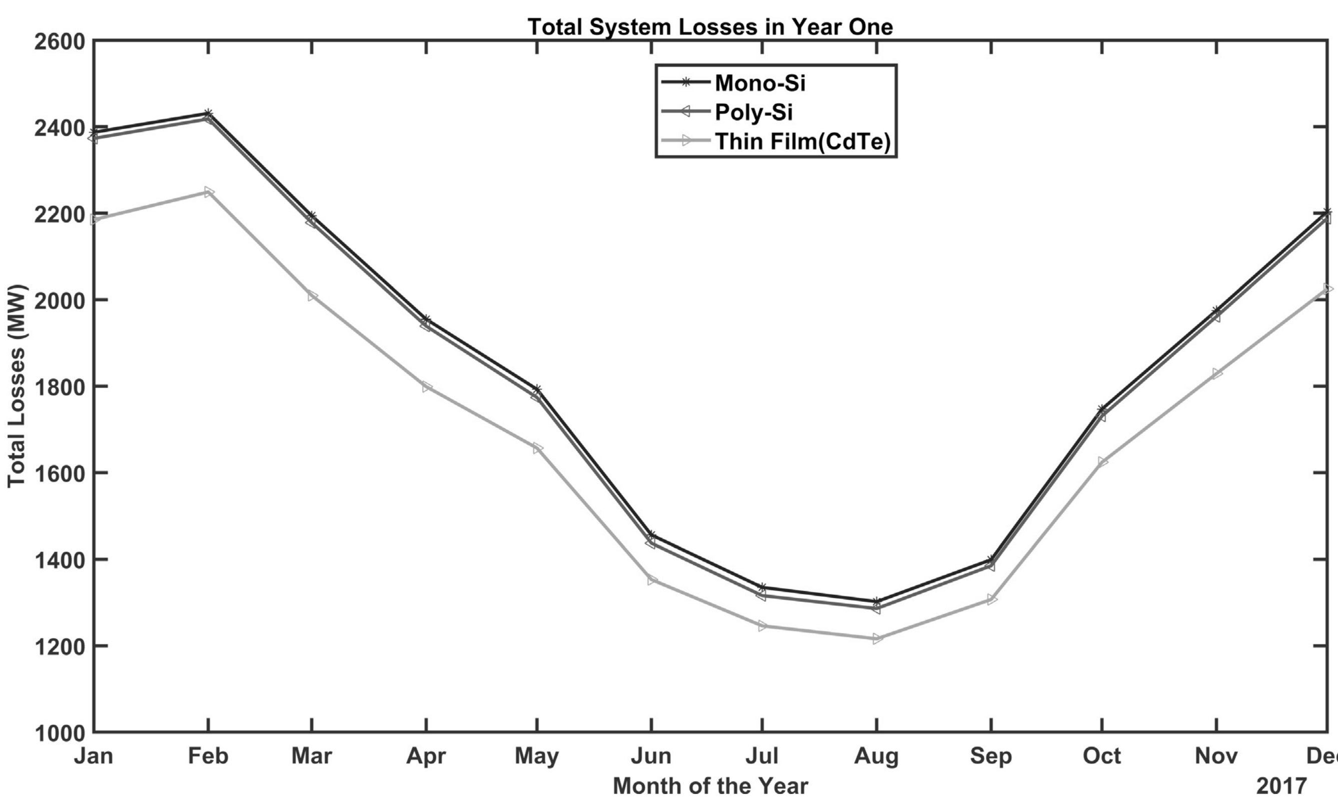

Fig. 9. Comparison of average monthly losses of the systems.

costs of energy generation by mono-Si, poly-Si, and CdTe systems were estimated as 12.4 cents/kWh, 12.3 cents/kWh, and 10.9 cents/kWh, respectively. The estimated costs of energy generation for all three systems are competitive because they are lower than the set feed-tariff of 14 cents/kWh in Ghana.

The estimated payback periods for mono-Si, poly-Si, and thin

Table 3
Performance ratio and Capacitor factors of selected solar PV installations.

Location	PV Technology	Performance ratio (%)	Capacity Factor (%)	Source
KNUST, Ghana	Mono-Si	67.9	11.47	Quansah et al. (2017)
	Poly-Si	76.3	12.9	
	a-Si	75.8	12.8	
	HIT	74.8	12.6	
Meknes, Morocco	Poly-Si	81.7		Allouhi et al. (2016)
Meknes, Morocco	Mono-Si	79.6		
NMMU South Africa	Poly-Si	84.3		Okello et al. (2015)
UENR Nsoatre Campus, Ghana	Mono-Si	75.5	15.37	Present Study
	Poly-Si	75.7	15.41	
	CdTe	77.4	15.75	

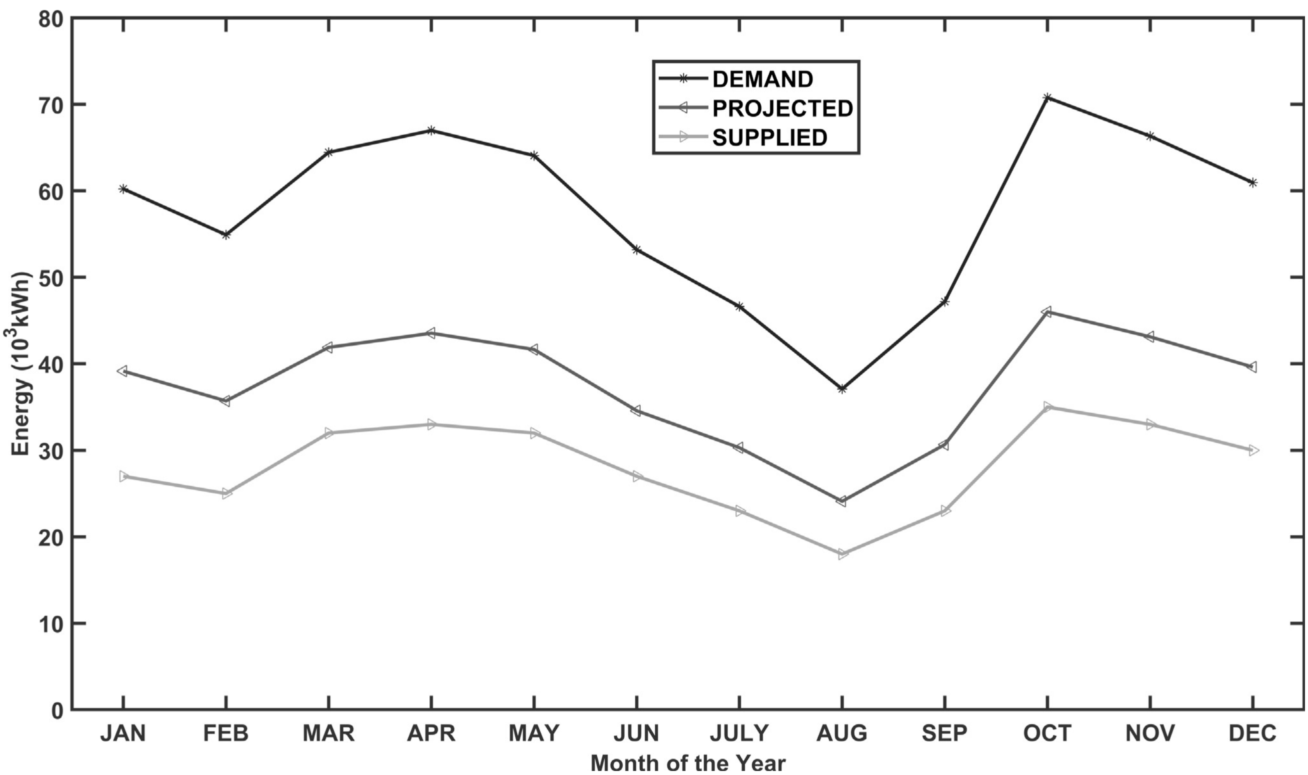

Fig. 10. Monthly average demand and supply.

Table 4
Summary of economic indicators of the PV systems.

PV System	NPV ($million)	Generation Cost (cents/kWh)	Cost-Benefit Ratio	PBP (Years)	After TAX-IRR
Mono-Si	6.911	12.4	1.25	7.2	3.8
Poly-Si	6.551	12.3	1.24	7.2	3.8
Thin Film (CdTe)	13.235	10.9	1.50	6.4	5.2

film systems are 7.2, 7.2, and 6.4 years, respectively. Overall, the thin film technology system presented higher net present value (NPV) and cost-benefit compared to the crystalline technology systems. Polycrystalline technology presented a slightly higher overall economic performance compared to monocrystalline PV system. The lower overall economic performance realized by monocrystalline technology is attributed to the high overall system costs, as mentioned earlier.

It can be observed from the presented results that all PV system options have positive economic evaluations results and, therefore, pass the screening criteria for project feasibility. The thin film PV system provides greater economic benefits compared to the crystalline silicon systems under nearly all analysed scenarios. This result can be attributed to the different losses and the corresponding economic benefits from increased energy sales.

4.4.2. *Sensitivity analysis*

A sensitivity analysis was performed to investigate the influence of key economic parameters on NPV and the cost of energy produced.

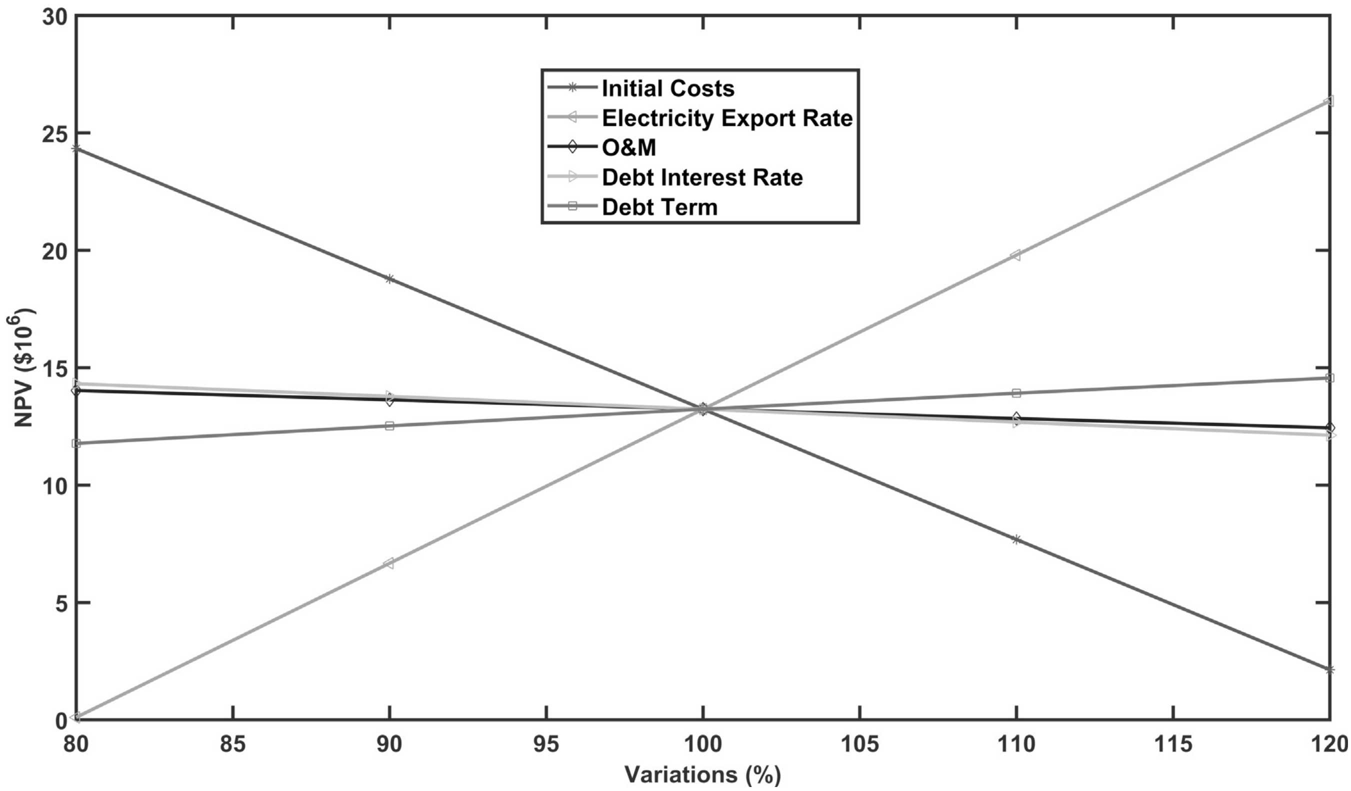

Fig. 11. Impacts of the Variations key parameters on NPV.

4.4.3. Net present value (NPV)

Fig. 11 shows the impact of variations of a few key parameters on the NPV of the system. It is clear from Fig. 11 that a decrease in the total system initial costs results in an increased NPV. For instance, a 20% reduction would cause NPV to increase from $13.2 million (reference case) to $24.3 million. An increase in the initial cost, in turn, results in a decrease in NPV. Compared to the reference NPV of $13.2 million, a 20% increase in initial costs results in approximately 83% decrease in NPV. However, it is important to note that despite the significant NPV reduction (83%) upon increased initial cost (20%), the value still remains positive and the investment is justified. Additionally, a 20% debt interest rate and operation and maintenance (O&M) increments result in a decreased NPV. However, the impact of debt interest rate and O&M variations on NPV were significantly lower compared to variations on initial costs.

Electricity export rate and debt term presented the same trend regarding the impact of their variation on NPV. However, the impact of electricity export rate is significantly higher than that of debt term, as shown in Fig. 11. A decrease in electricity export rate results in a decreased NPV, whereas an increase results in an

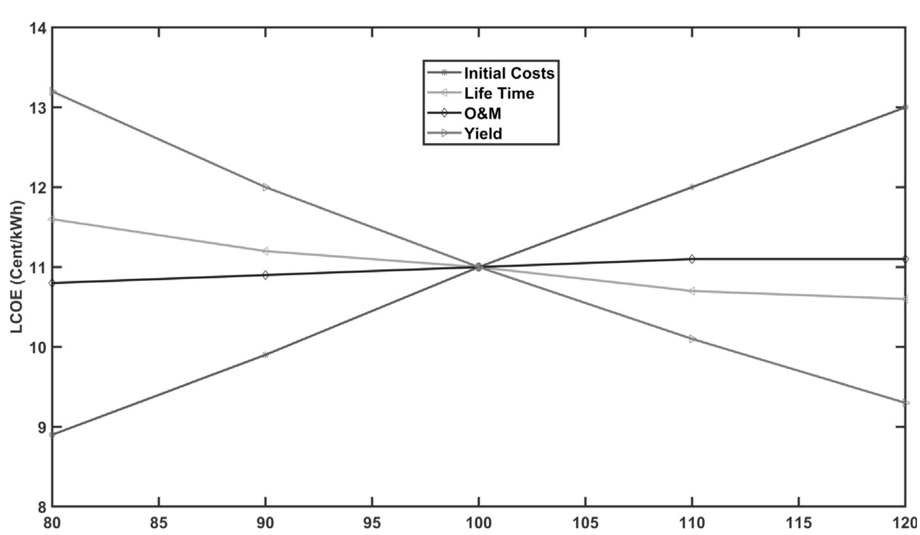

Fig. 12. Impacts of variation of key parameters on cost of energy.

increased NPV. The two most important parameters on the NPV variations are initial costs and electricity export rate.

4.4.4. Energy production cost

Decreased initial costs result in decreased cost of energy produced, whereas an increase results in increased energy costs. In contrast, decreased yield, O&M, and lifetime of the system leads to an increased energy generation cost, whereas an increase results in decreased cost of energy generation. The impact of yield variation is significantly higher than that of O&M and lifetime, though they followed the same trend. Moreover, the significant parameters affecting the cost of energy produced are initial costs and the energy yield, as shown in Fig. 12.

5. Conclusion

The suitability of the selected location for the implementation of a solar photovoltaic power plant was adequately assessed. From the results of the site assessments, there were no barriers for the development of the project. The thin film technology system emerged as the best PV technology, with 77.5% PR. It was followed by polycrystalline and monocrystalline silicon systems with 75.7% and 75.5% PR, respectively. The costs of the energy produced by the monocrystalline, polycrystalline, and thin film systems are 12.4, 12.3, and 10.9 cents/kWh, respectively. The costs of energy determined for these systems are lower than the set FiT of 14 cents/kWh.

The sensitivity analyses showed that the most relevant parameters that significantly affect PV power generation costs, NPV, and PB are the initial project costs, electricity export rate, and energy yields of the systems.

Based on these findings, the development of a 50-MW grid-tied solar photovoltaic plant at UENR Nsoatre campus is both technically and economically feasible. The proposed plant has the capacity to supply more than 48% of the electricity demand on campus and fulfil a sizeable portion of the country's renewable energy generation targets. Moreover, the utilization of indigenous resources for the proposed utility scale plant would reduce the country's increasing dependency on energy imports, contribute to Ghana's energy security efforts, and reduce the University's carbon footprint significantly.

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